

When Not to Trust Your Model

Module 5 Week B – Trees, Ensembles & Model Selection

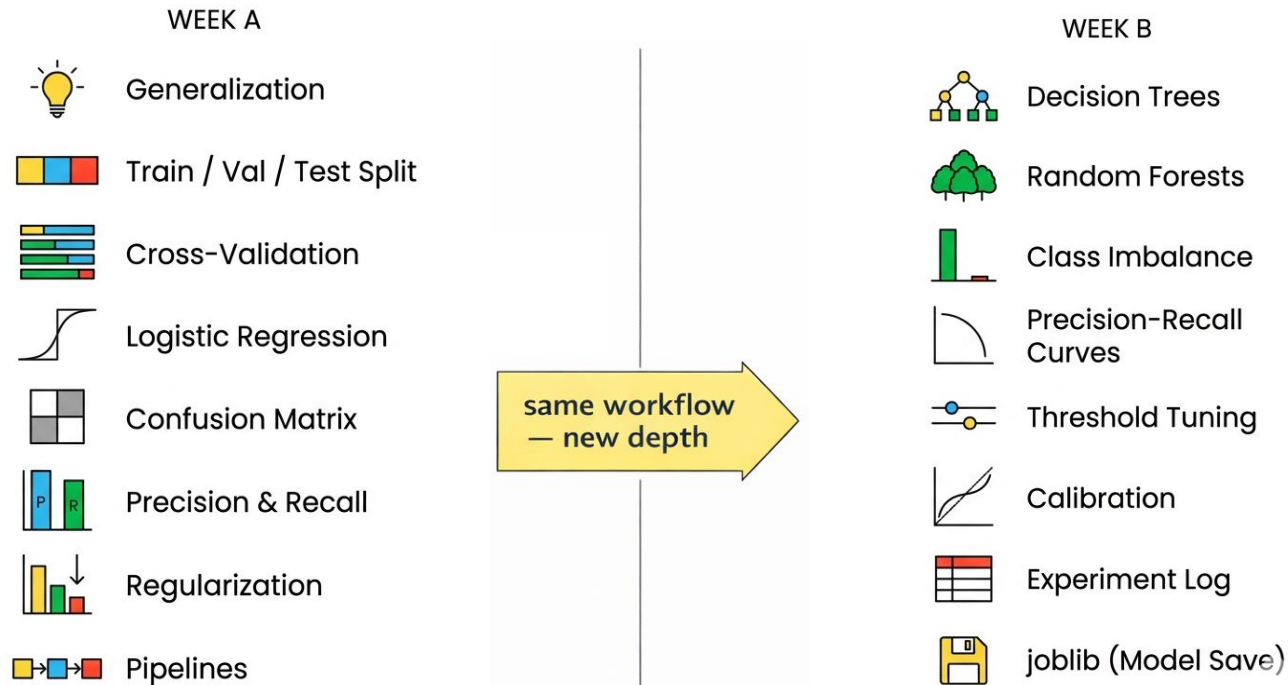
Get to know Lana — Fun Fact



I love live music

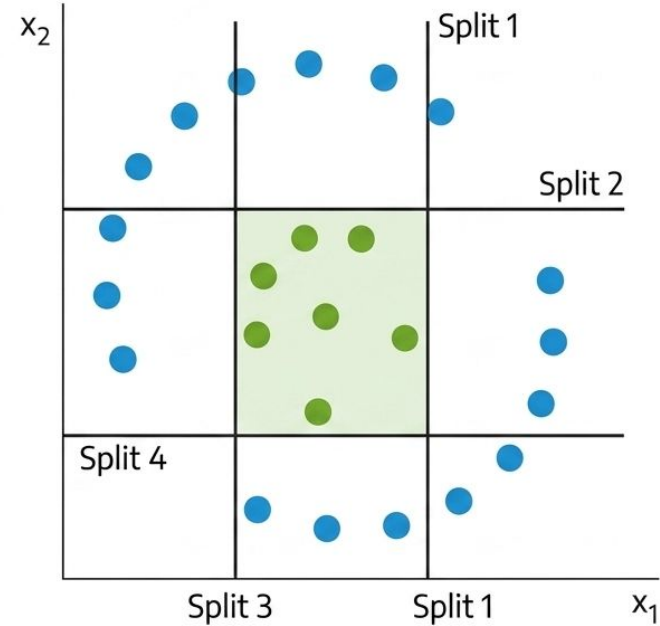
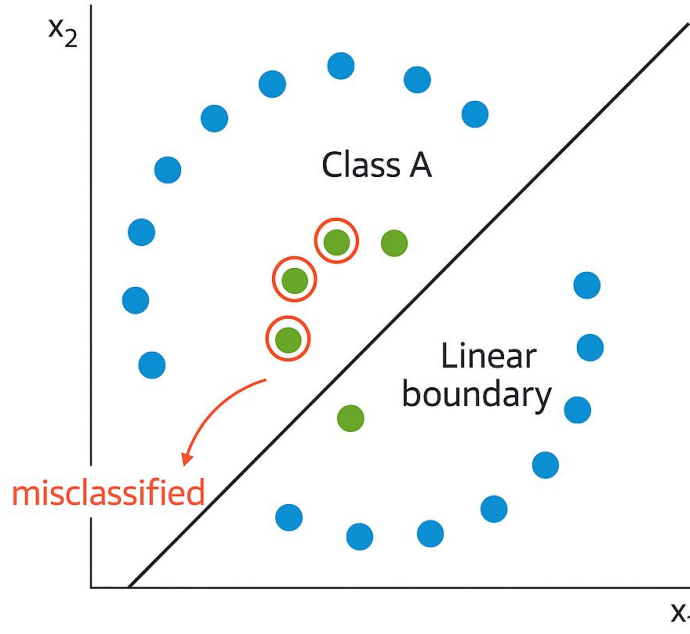
- From arenas to speakeasies
- All genres, always exploring
- Energy, dancing, relaxing
- Some of my best memories start here

Module 5: Last Week to This Week



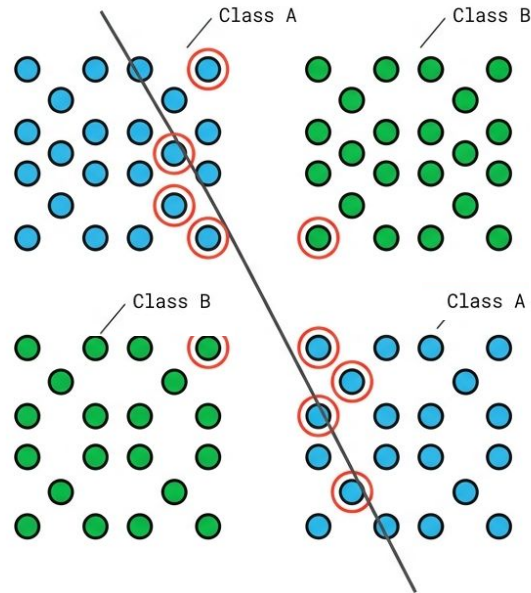
Beyond Straight Lines

Linear Boundaries

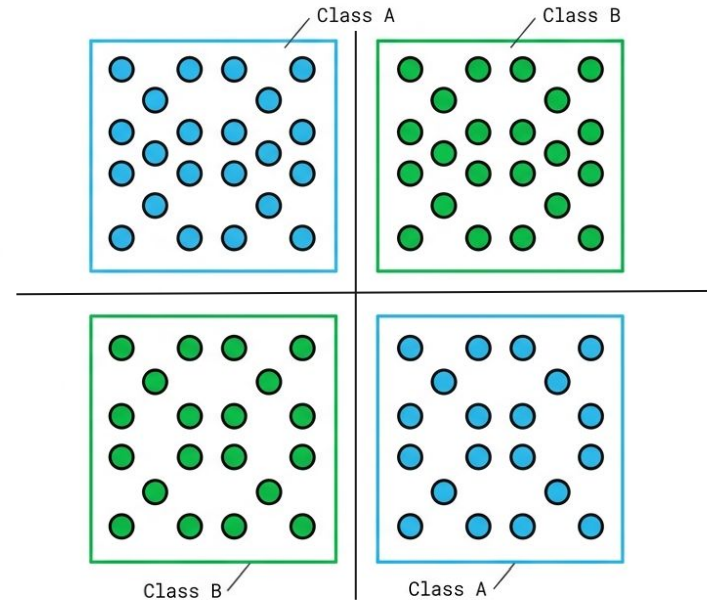


Tree Partitioning

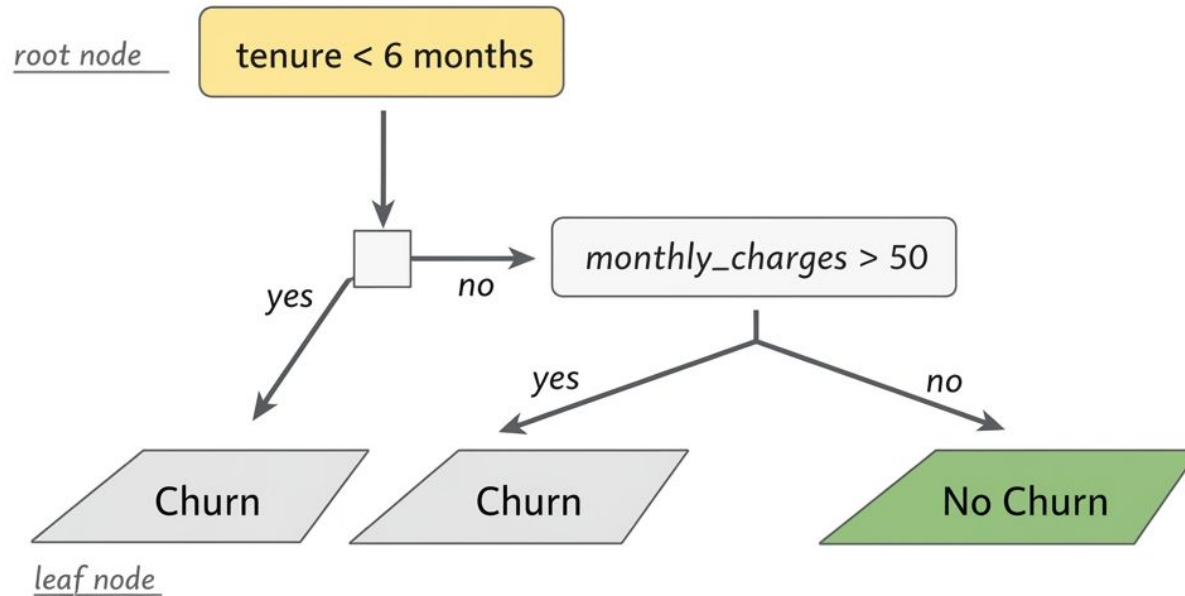
Line-based separator



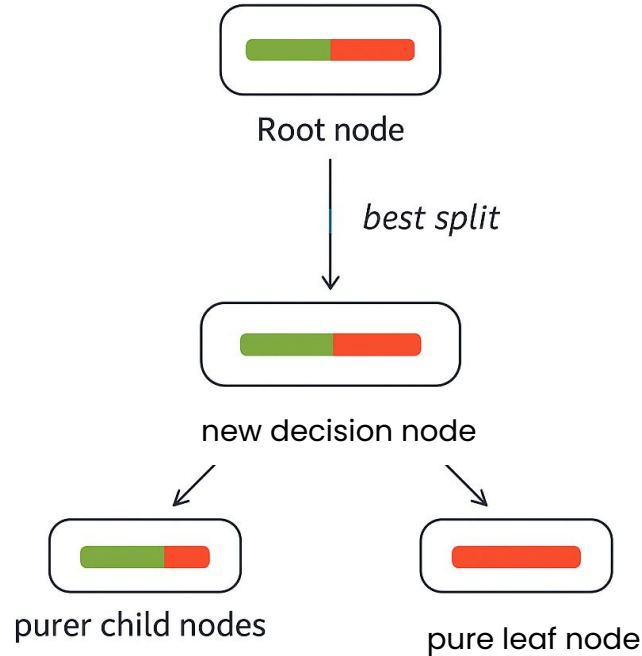
Axis-aligned rectangular partitions



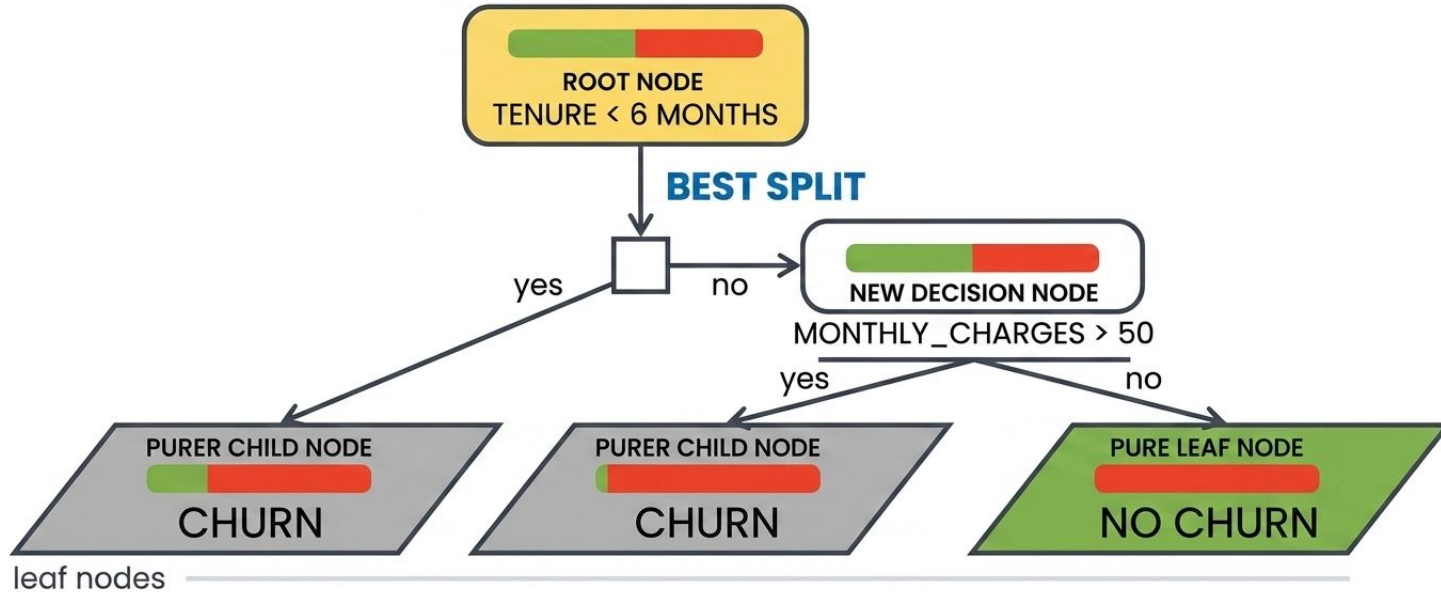
How a Tree Grows



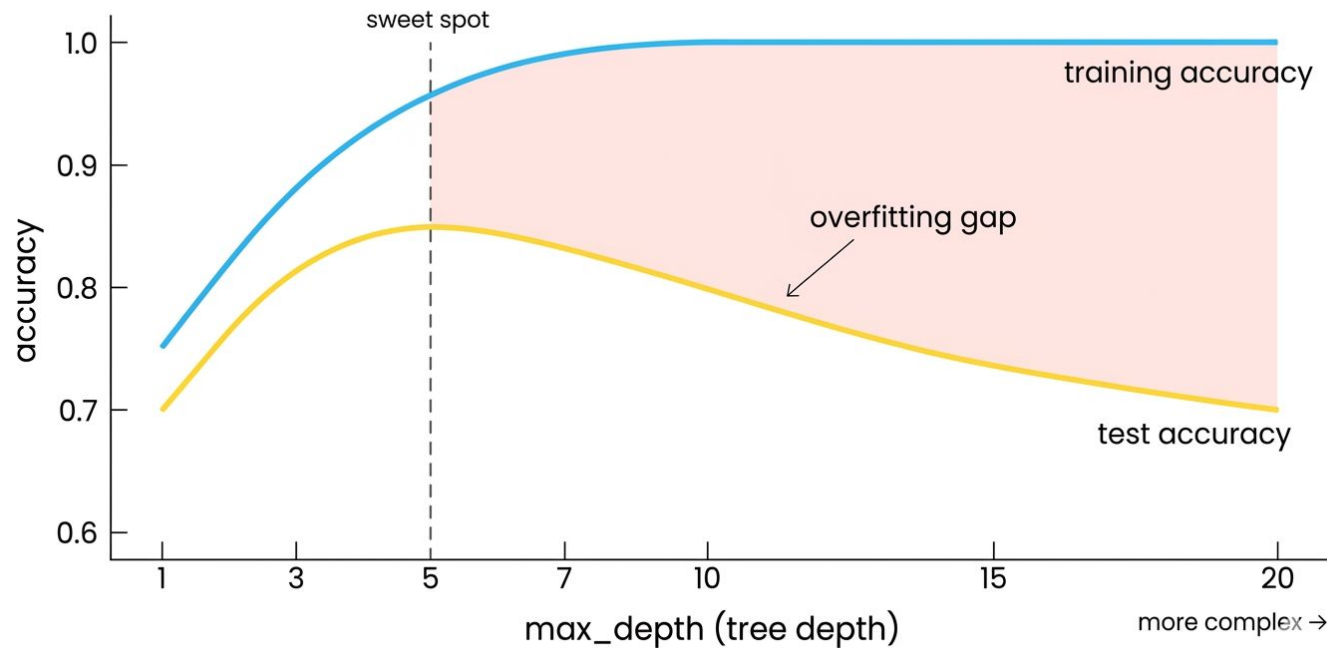
Clean Splits, Pure Leaves



The Tree in our Example



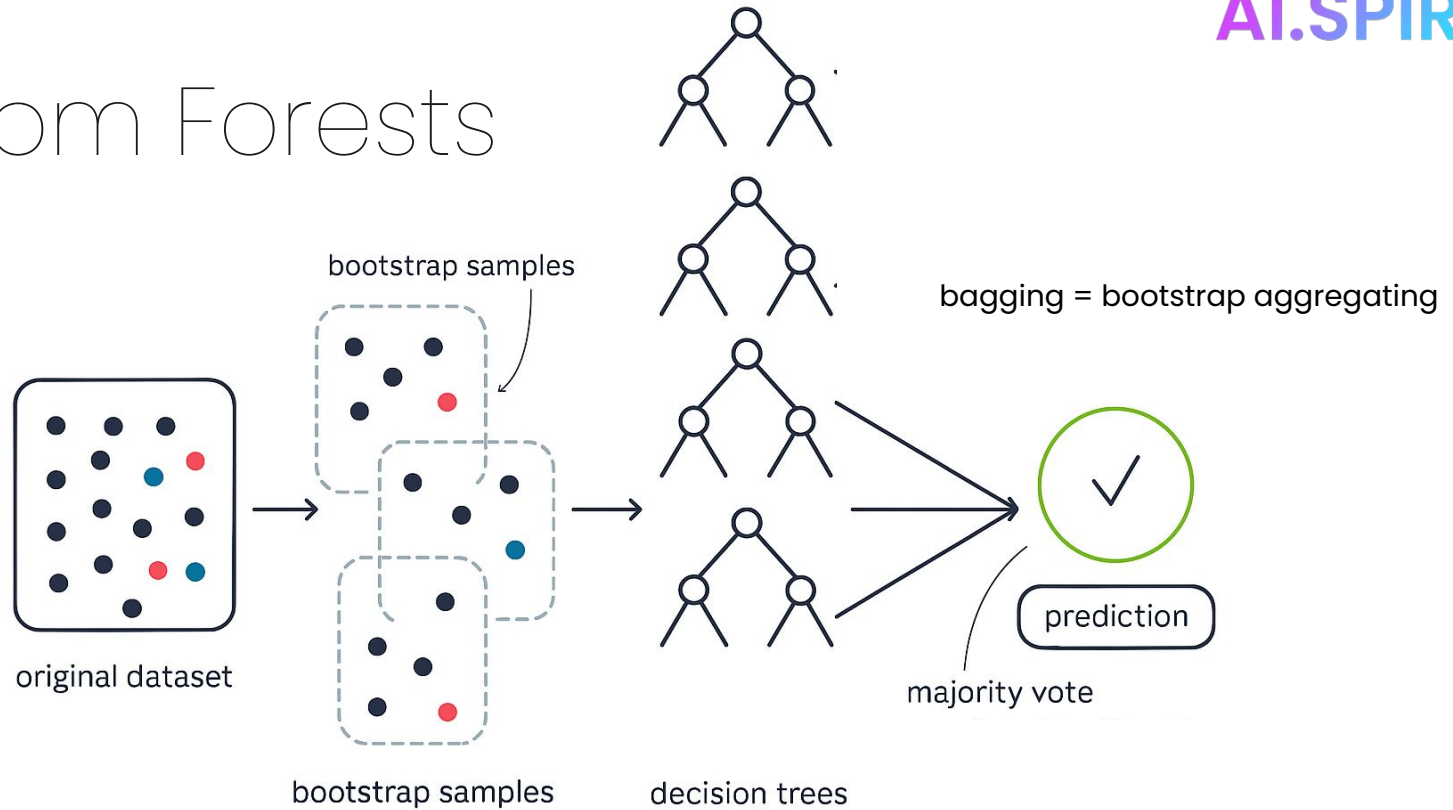
Overfitting a Tree



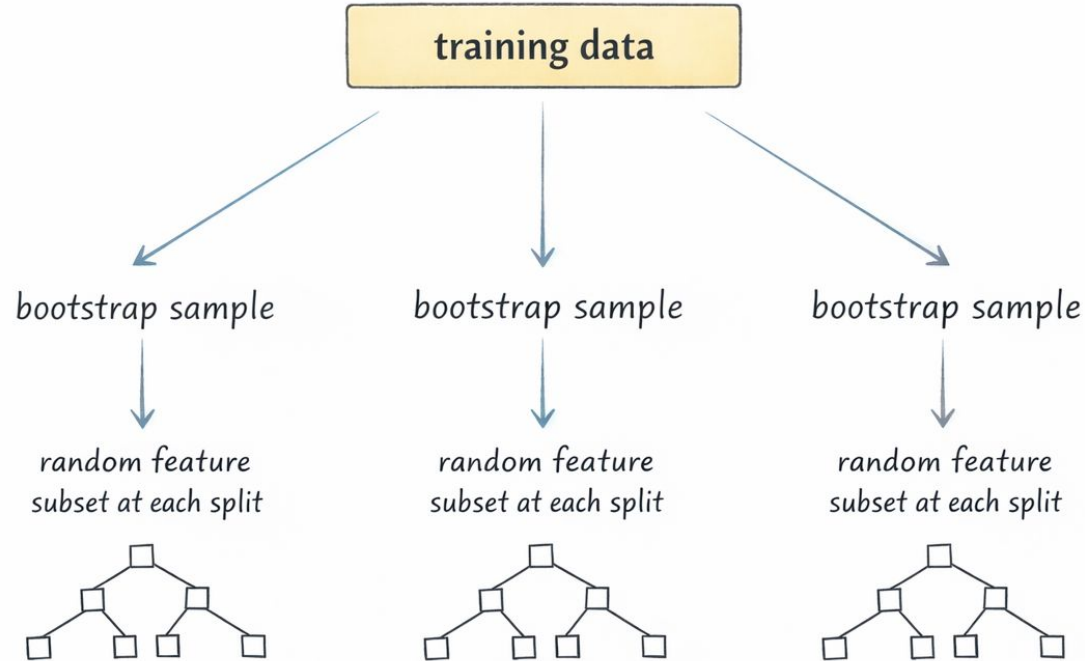
Tree Variance



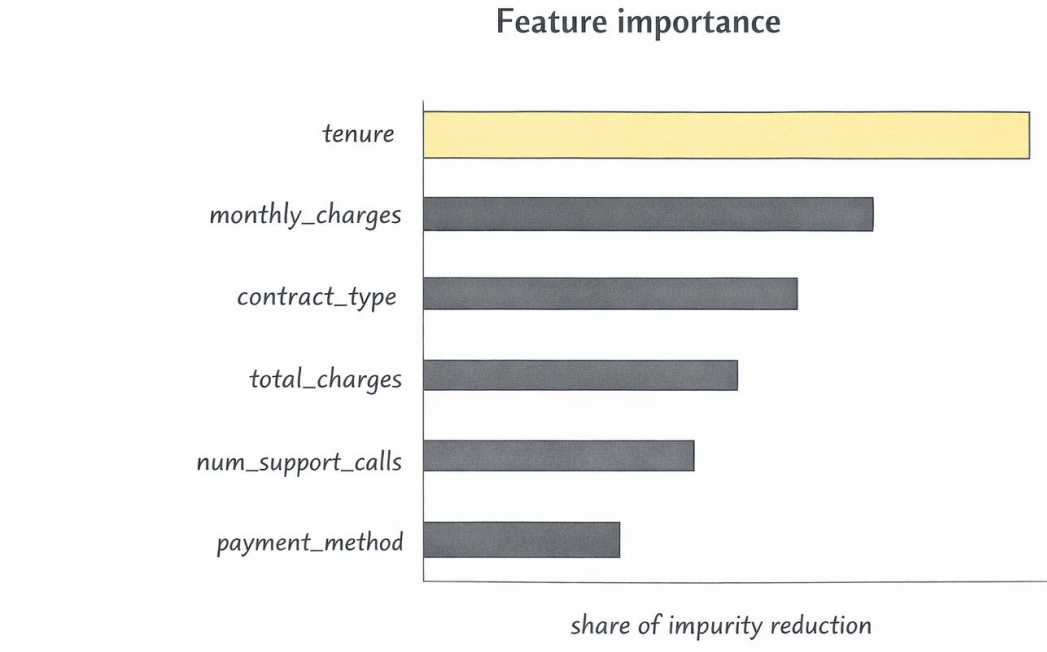
Random Forests



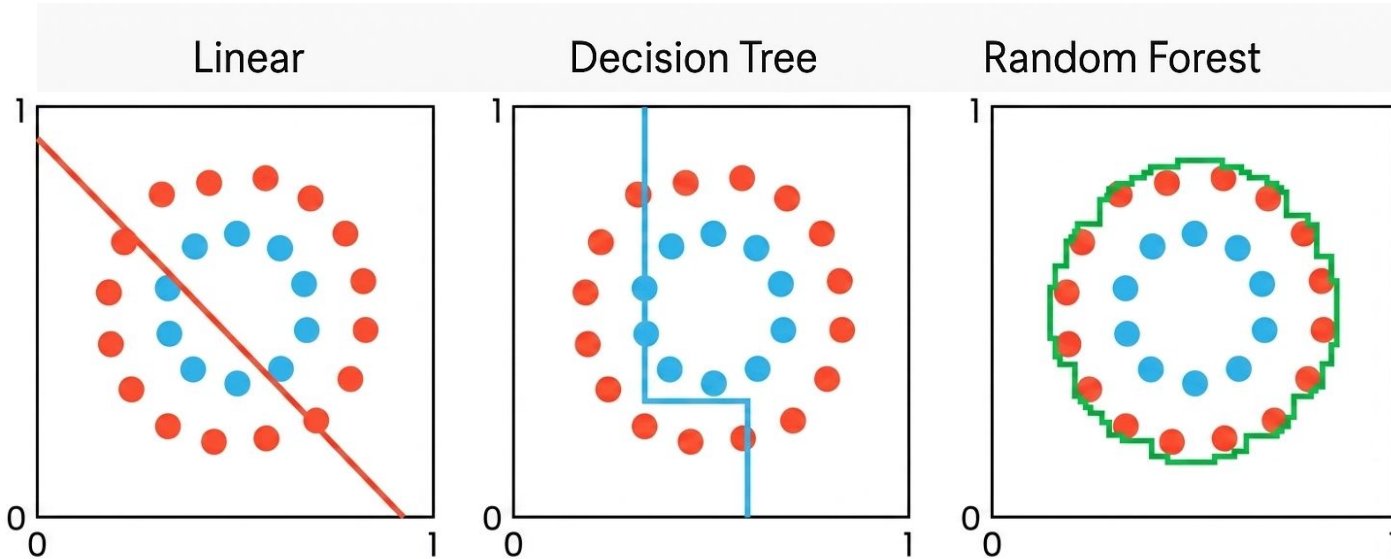
Two Sources of Randomness



Which Features Matter



Three Models, Three Boundaries



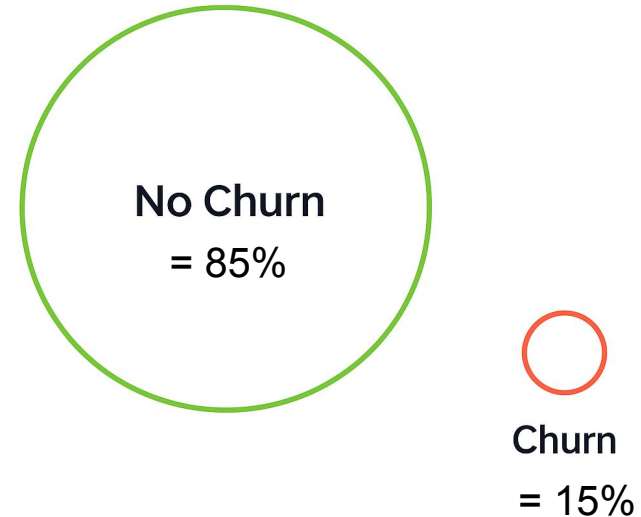
Imbalance

The Accuracy Trap

Last week



This week



Reading the Confusion Matrix

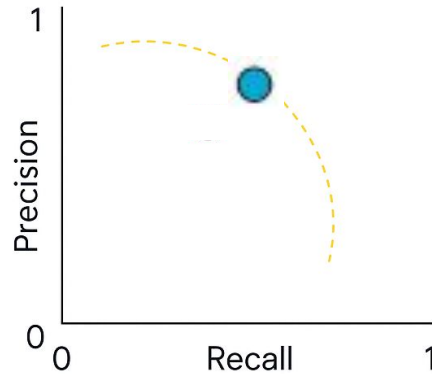
		Predicted	
		Negative	Positive
Actual	Negative	TN 850	FP 0
	Positive	FN 150	TP 0

model never predicts positive ↗

Positive class completely missed → recall = 0

Class Weights

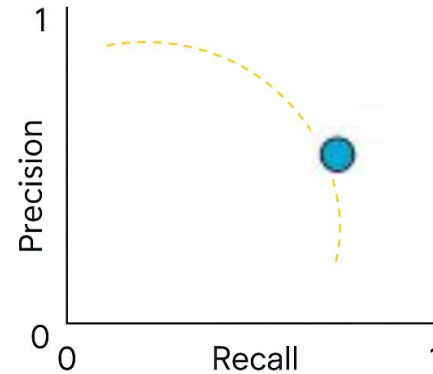
Equal class weight



Recall  0.50

Precision  0.75

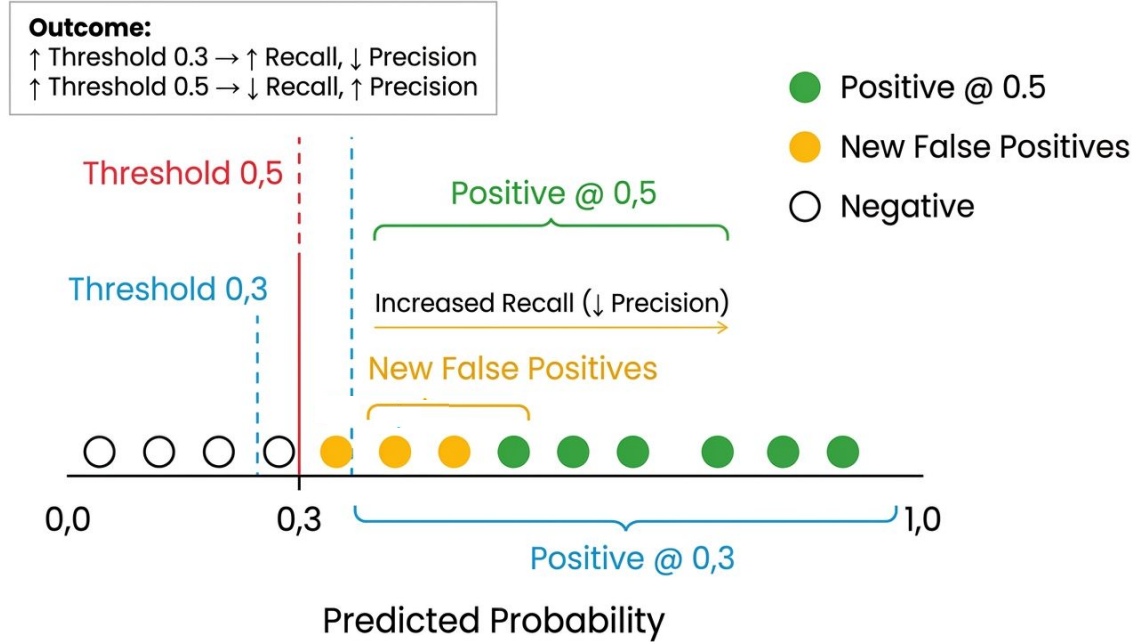
Minority-weighted loss



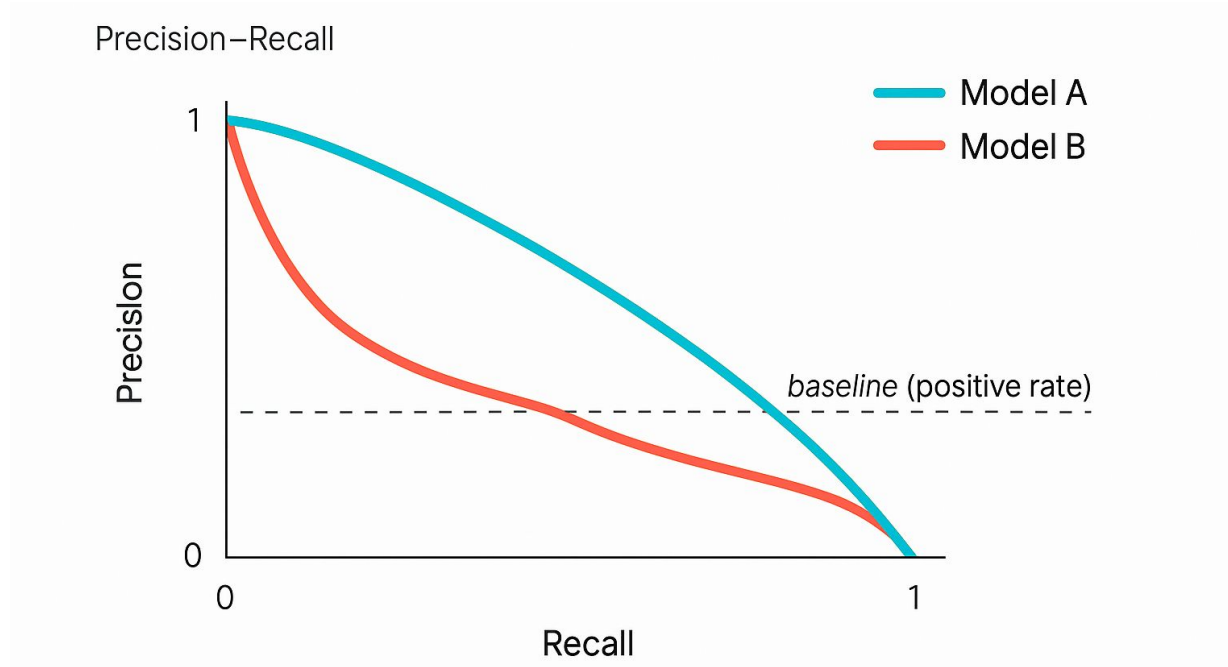
Recall  0.75

Precision  0.50

Threshold Tuning

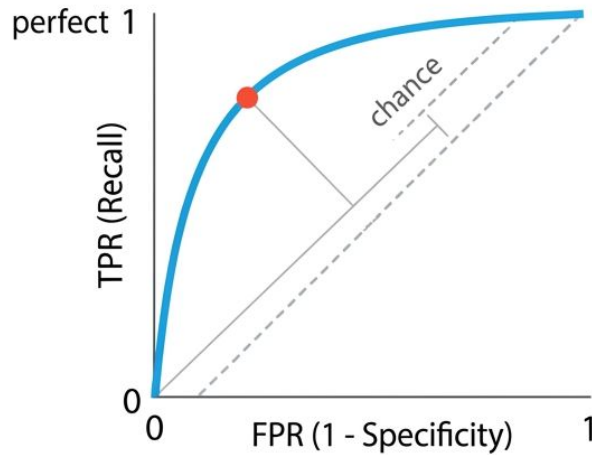


Precision vs Recall

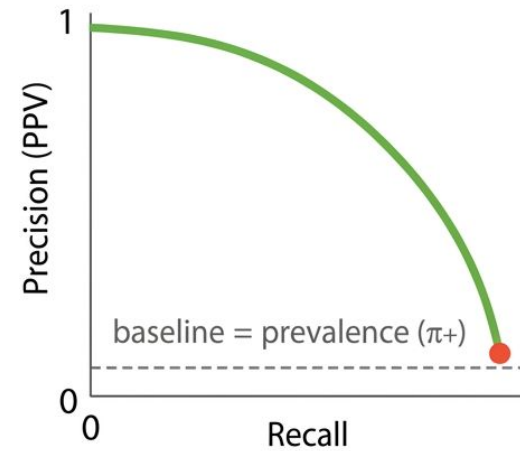


PR-AUC

ROC framing



PR framing (positive-class focus)

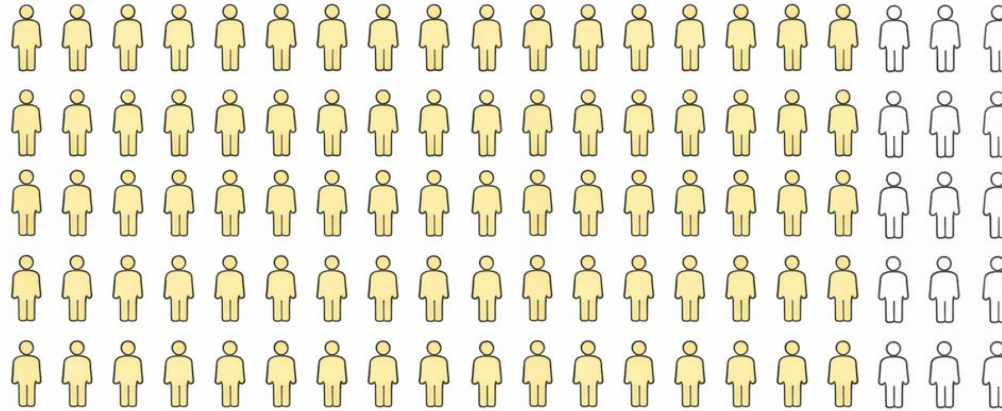


average_precision_score

Trust but Verify

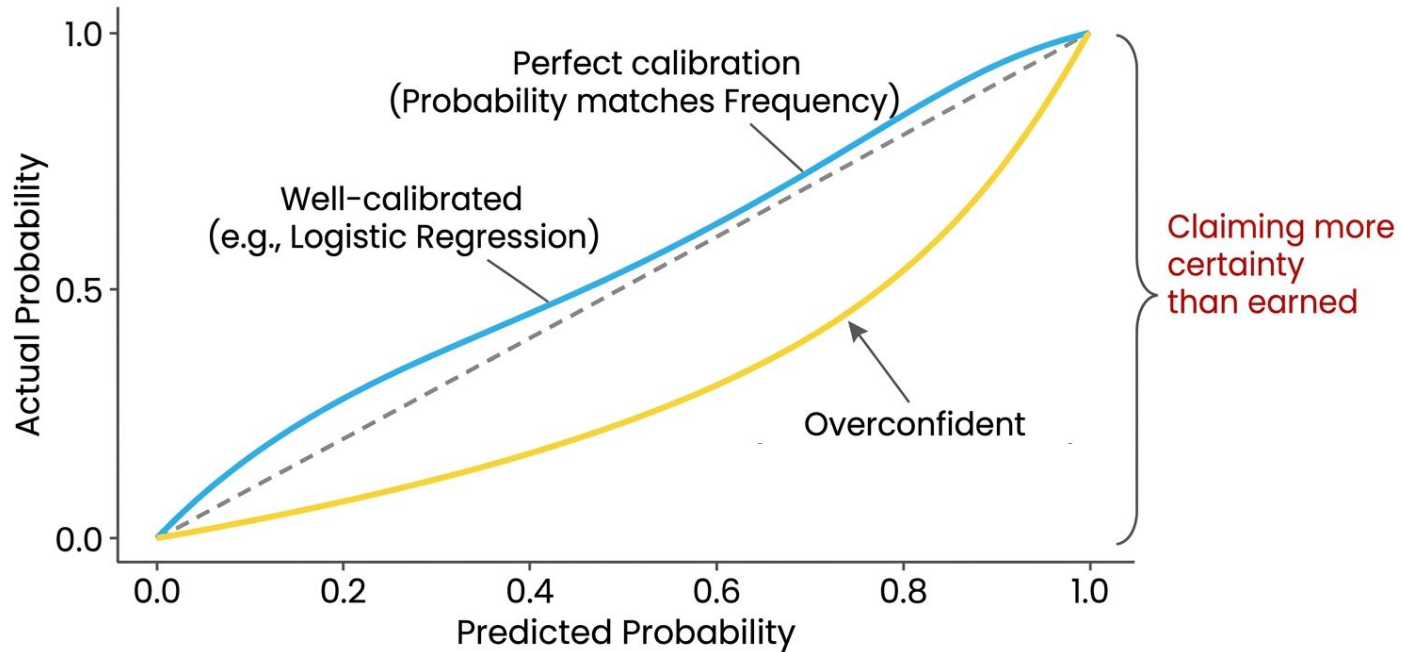
Calibration

model score 0.85

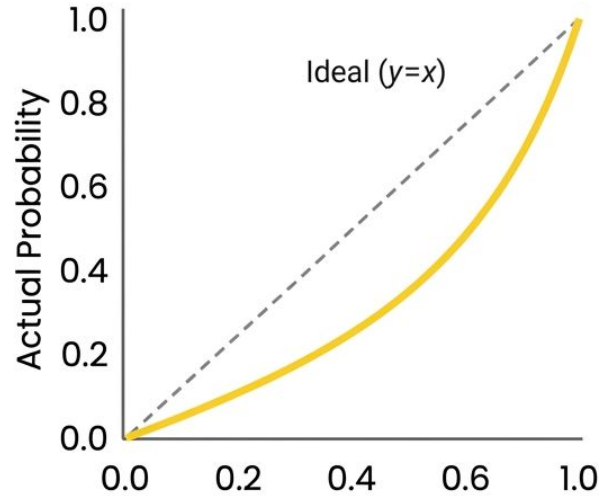


of customers given score 0.85, about 85% should actually churn

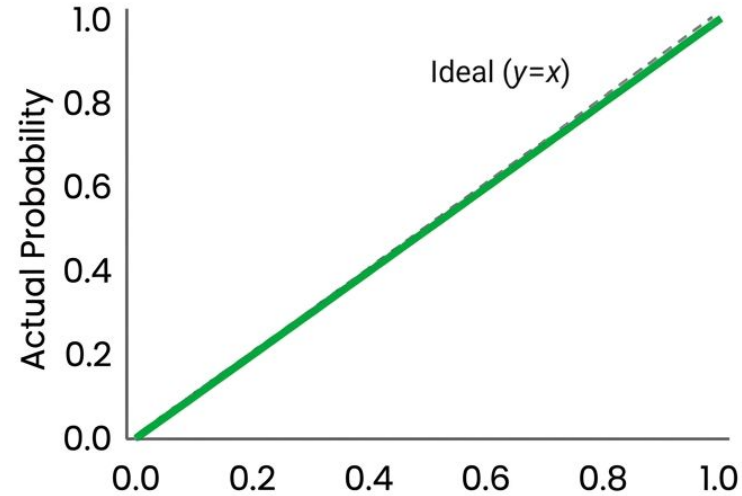
Calibration: Can You Trust the Score?



Calibrating Probabilities



Before calibration
(Random Forest)



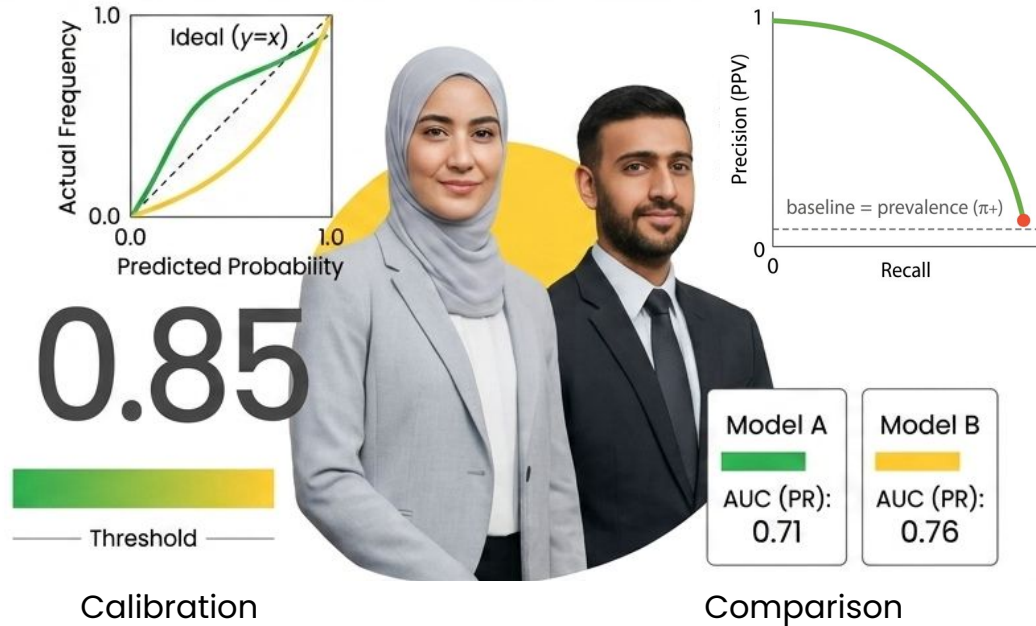
After calibration

Choosing a Model

Factors Beyond Scores

Criteria	Linear Models	Tree Ensembles
Interpretability	 HIGH	 LOW
Flexibility	 LOW-MED	 HIGH
Calibration	 HIGH	 MED
Complexity	 LOW	 HIGH

Diagnosing the 85% Model



The Reproducible Workflow



Questions

Announcements

Congratulations!



This week's Module Mentors

- Malak
- Luma
- Ameer